Physics 5313: Electromagnetic Theory 2 (Fall 2026)

Lecture 15: Resonant Cavity 2

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Power loss in a cavity

- Definition of Q

WG: $k = k_0 + \alpha + i\beta$

Perfect conductor: resonant freq: ω_0

Good conductor: $\omega = \omega_0 + \Delta\omega - i\omega_i$

U is total EM energy inside the resonant cavity.

Time dependence of field E

$$\vec{E}(\vec{x},t) = \vec{E}_0(\vec{x})\, e^{-i\omega t}$$

$$= \vec{E}_0\, e^{-i\omega_0 t}\, e^{-i(\omega_0+\omega)t}$$

$$U = \int u \, dV = \int u \, dx\, dy\, dz$$

$$= \int \left(\frac{\varepsilon}{2}|\vec{E}|^2 + \frac{1}{2}|\vec{B}|^2\right) dx\, dy\, dz$$

$$\frac{1}{2}|\vec{B}|^2 = \frac{1}{2}|\vec{E}|^2$$

Time dependence of field E

$$U = \int \varepsilon |\vec{b}|^2 \, da \, dz$$

$$= \int \varepsilon |\vec{b}_0|^2 e^{-2\omega i t} \, da \, dz$$

$$= \varepsilon e^{-2\omega i t} \int |\vec{b}_0(\vec{x})|^2 \, da \, dz$$

$$= U_0 e^{-2\omega i t}$$

Time dependence of field E

$$U = U_0 e^{-2\omega_i t}$$

$$\frac{dU}{dt} = P_{loss} = -2\omega_i U_0 e^{-2\omega_i t} = -2\omega_i U$$

$$\Rightarrow \quad -2\omega_i = -\frac{P_{loss}}{U} = \frac{dU/dt}{U}$$

$$\omega_i = \frac{1}{2}\frac{P_{loss}}{U}$$

Time dependence of field E

Define $Q = \frac{\omega_0}{2\omega_i}$

Q : how many periods can the EM waves last in the resonant cavity.

$$U = U_0 e^{-\frac{\omega_0}{Q} t}$$

Time dependence of field E

$$\vec{E}(\vec{x},\omega) = \frac{1}{\sqrt{2\pi}} \int_0^\infty \vec{E}(\vec{x},t)\, e^{i\omega t}\, dt$$

$$= \frac{1}{\sqrt{2\pi}} \int_0^\infty \vec{E}_0(\vec{x})\, e^{-\frac{\omega_0 t}{2\theta}}\, e^{-i(\omega_0+\Delta\omega)t}\, e^{i\omega t}\, dt$$

$$= \frac{1}{\sqrt{2\pi}} \frac{\vec{E}_0(\vec{x})}{\frac{\omega_0}{2\theta} - i\omega + i\omega_0 + i\Delta\omega}$$

$$I(\omega) \propto |\vec{E}(\omega)|^2 \propto \frac{1}{\left(\frac{\omega_0}{2\theta}\right)^2 + (\omega - \omega_0 - \Delta\omega)^2}$$

Time dependence of field E

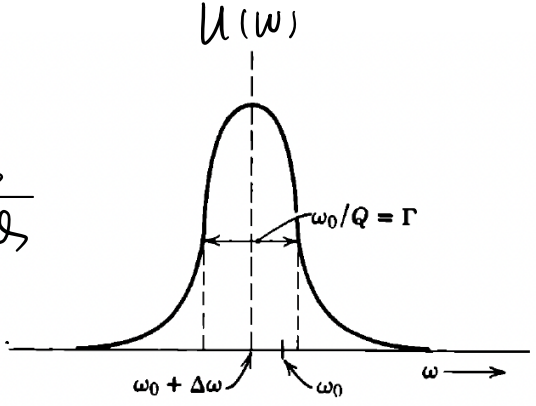
$$U(\omega) \propto |\vec{b}(\omega)|^2 \propto \frac{1}{\left(\frac{\omega_0}{2Q}\right)^2 + (\omega - \omega_0 - \Delta\omega)^2}$$

Max at $\omega = \omega_0 + \Delta\omega$

Half max at $\omega = \omega_0 + \Delta\omega \pm \frac{\omega_0}{2Q}$

T : full width of half max.

$$T = \frac{\omega_0}{Q}$$



Example: Energy U in cavity

TM mode: $H_z = 0$

$$U = \int u \left|\vec{H_t}\right|^2 da\, dz$$

$$\langle U \rangle = \int \frac{1}{2} u \left|\vec{H_t}\right|^2 da\, dz \qquad \text{time-averaged } U$$

$$\vec{H_t} = \frac{i\xi w}{\gamma^2} \cos\left(\frac{p\pi z}{d}\right) \vec{z} \times \nabla_t \psi$$

$$\left|\vec{H_t}\right|^2 = \frac{\xi^2 w^2}{\gamma^4} \cos^2\left(\frac{p\pi z}{d}\right) \left|\nabla_t \psi\right|^2$$

Example: Energy U in cavity

$$\langle U \rangle = \int \frac{1}{2} u \frac{\mathcal{E}^2 w^2}{\gamma 4} \cos^2\left(\frac{p\pi z}{d}\right) |\nabla_t \psi|^2 \, da \, dz$$

$$= \frac{1}{2} \frac{u \mathcal{E}^2 w^2}{\gamma 4} \frac{d}{2} \int |\nabla_t \psi|^2 \, da \qquad (p > 0)$$

$$= \frac{d}{4} \frac{u \mathcal{E}^2 w^2}{\gamma 4} \frac{\int_A |\nabla_t \psi|^2 \, da}{\gamma^2 \int_A |\psi|^2 \, da} \qquad \text{Green's 1st}$$

Example: Energy U in cavity

$$\langle U \rangle = \frac{d}{4} \frac{\mu \varepsilon^2 \omega^2}{\gamma^2} \int_A |\psi|^2 \, da$$

Because $\mu \varepsilon \omega^2 = \gamma^2 + k^2$

$$= \frac{d}{4} \varepsilon \frac{\gamma^2 + k^2}{\gamma^2} \int_A |\psi|^2 \, da$$

$$= \frac{d}{4} \varepsilon \left| 1 + \left(\frac{P\pi}{d\gamma} \right|^2 \right] \int_A |\psi|^2 \, da$$

TE: $$\langle U \rangle = \frac{d}{4} \mu \left| 1 + \left(\frac{P\pi}{d\gamma} \right|^2 \right] \int_A |\psi|^2 \, da$$

Power loss in a cavity

- Power loss P

$$\frac{dP_{loss}}{da} = \frac{\mu_c \delta w}{4} |\vec{H}_t|^2$$

$$P_{loss} = \int \frac{dP_{loss}}{da} da$$

$$= \frac{1}{2\delta S} \left[\oint dL \int_0^d dz \; |\vec{n} \times \vec{H}|^2_{sides} + 2 \oint da \, |\vec{n} \times \vec{H}|^2_{ends} \right]$$

Power loss in a cavity

- Power loss P

Assume TM mode

$$\vec{H} = \vec{H}_t = \frac{i\varepsilon\omega}{\gamma^2}\cos\left(\frac{p\pi}{d}z\right)\hat{z}\times\nabla_t\psi$$

Sides: $\left|\vec{n}\times\vec{H}\right|^2 = \left|\frac{\partial\psi}{\partial n}\right|^2\frac{\varepsilon^2\omega^2}{\gamma^4}\cos^2\left(\frac{p\pi}{d}z\right)$ $\quad p>0$

$$P_{loss,1} = \frac{1}{2\delta s}\int\frac{d}{2}\oint dl\left|\frac{\partial\psi}{\partial n}\right|^2\frac{\varepsilon^2\omega^2}{\gamma^4}$$

$$= \frac{d}{4\delta s}\frac{\varepsilon^2\omega^2}{\gamma^4}\oint\left|\frac{\partial\psi}{\partial n}\right|^2 dl$$

Power loss in a cavity

Ends: $|\vec{n} \times \vec{H}|^2 = \frac{\epsilon^2 w^2}{\gamma^4} |\vec{n} \times (\hat{z} \times \nabla_\perp \psi)|^2$

$\vec{n} = \pm \hat{z}$

$= \frac{\epsilon^2 w^2}{\gamma^4} |\nabla_\perp \psi|^2$

$P_{loss,2} = \frac{1}{\delta b} \frac{\epsilon^2 w^2}{\gamma^4} \int |\nabla_\perp \psi|^2 da$

$= \frac{1}{\delta b} \frac{\epsilon^2 w^2}{\gamma^2} \int |\psi|^2 da$ Green's 1st

Power loss in a cavity

$$P_{loss} = \frac{d}{4\delta}\frac{\varepsilon^2\omega^2}{\sigma^4}\oint\left|\frac{\partial\psi}{\partial n}\right|^2 dl + \frac{1}{\delta\sigma}\frac{\varepsilon^2\omega^2}{\sigma^2}\int|\psi|^2 da$$

$$= \frac{\varepsilon^2\omega^2}{\delta\sigma^4}\left(\frac{d}{4}\oint\left|\frac{\partial\psi}{\partial n}\right|^2 dl + \sigma^2\int|\psi|^2 da\right)$$

Power loss in a cavity

C: Circumference.

A: Cross section area

Use Eq 8.62

$$\oint \left|\frac{\partial \psi}{\partial n}\right|^2 dl = \varepsilon_\lambda \gamma^2 \frac{C}{A} \int_A |\psi|^2 da$$

$$\therefore P_{loss} = \frac{\varepsilon^2 w^2}{8\delta\gamma^4} \left(\frac{d}{4} \varepsilon_\lambda \gamma^2 \frac{C}{A} \int_A |\psi|^2 da + \gamma^2 \int |\psi|^2 da \right)$$

$$= \frac{\varepsilon^2 w^2}{8\delta\gamma^2} \left(1 + \frac{Cd}{4A} \varepsilon_\lambda\right) \int_A |\psi|^2 da$$

Power loss in a cavity

$$w_i = \frac{1}{2}\frac{P_{loss}}{U}$$

$$Q = \frac{w_0}{2w_i}$$

$$Q = \frac{w_0 U}{P_{loss}}$$

$$= \frac{w_0 \frac{d}{4}\varepsilon\left[1+\left(\frac{P\bar{\pi}}{d\gamma}\right)^2\right]\int_A |\psi|^2 da}{\frac{\varepsilon^2 w^2}{8b\gamma^2}\left(1+\frac{Cd}{4A}\varrho_\lambda\right)\int_A |\psi|^2 da}$$

$$= \frac{w_0 \frac{d}{4} u}{\frac{1}{8b}\left(1+\frac{Cd}{4A}\varrho_\lambda\right)}$$

Power loss in a cavity

$$Q_s = \frac{w_0 \frac{d}{4} u}{\frac{1}{\delta b}\left(1 + \frac{cd}{4A} \ell_{n\lambda}\right)}$$

$$b = \frac{2}{u_c w \delta^2} \qquad Q_s = \frac{w_0 \frac{d}{4} \frac{2u}{u_c w \delta}}{1 + \frac{cd}{4A} \ell_{n\lambda}} \qquad \text{Let } w_0 = w$$

$$Q_s = \frac{u}{u_c} \frac{d}{\delta} \frac{1}{1 + \frac{cd}{4A} \ell_{n\lambda}}$$

Power loss in a cavity

$$Q_s = \frac{\mu}{\mu_c}\frac{d}{\delta}\frac{1}{2\left(1+\xi_1\frac{Cd}{4A}\right)}$$

$V = dA$

S is total surface area

$$= \frac{\mu}{\mu_c}\frac{1}{\delta}\frac{dA}{2\left(A+\frac{\xi_1 Cd}{4}\right)}$$

$$= \frac{\mu}{\mu_c}\frac{V}{S\delta} \times \text{Geometrical Factor}$$

Perturbation of boundary conditions

WG: $\gamma_0^2 - \gamma^2 = k^2 - k^{(0)2} \simeq f \dfrac{\oint_C \left|\frac{\partial \psi_0}{\partial n}\right|^2 dl}{\int_A |\psi_0|^2 da}$ $\qquad f = (1+i)\frac{\kappa_c \delta}{2\kappa} \quad \frac{w^2}{w_0^2}$

In RC. $w = w_0$, $f = (1+i)\frac{\kappa_c \delta}{2\kappa}$

RC : $\rightarrow$ $w_0^2 - w^2 = (1+i)\frac{\kappa_c \delta}{2\kappa} \dfrac{\oint_C \left|\frac{\partial \psi_0}{\partial n}\right|^2 dl}{\int_A |\psi_0|^2 da}$

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Perturbation of boundary conditions

$$w_0^2 - w^2 = (1+i)\,\bar{I}$$

$$w^2 = w_0^2 - (1+i)\,\bar{I} = w_0^2\left(1 - \frac{1+i}{w_0^2}\,\bar{I}\right)$$

$$w = w_0\left(1 - \frac{1+i}{w_0^2}\,\bar{I}\right)^{1/2}$$

Assume $\bar{I} \ll w_0^2$

$$w_i = \frac{1}{2}\,\frac{P_{loss}}{|\mu|}$$

$$w = w_0\left(1 - \frac{1+i}{2w_0^2}\,\bar{I}\right) = w_0 - \frac{1+i}{2w_0}\,\bar{I}$$

$$Q = \frac{w_0}{2w_i}$$

$$= w_0 - \frac{\bar{I}}{2w_0} - i\,\frac{\bar{I}}{2w_0}$$

$$= w_0 + \delta w - i w_i$$

Perturbation of boundary conditions

$$\Delta w = -\frac{I}{2w_0}$$

$$\frac{I}{2w_0} = w_i = \frac{w_0}{2Q} \quad \Rightarrow \quad I = \frac{w_0^2}{Q}$$

$$\Delta w = -\frac{w_0^2}{Q} \times \frac{1}{2w_0}$$

$$= -\frac{w_0}{2Q} = -w_i$$

Resonant freq will be lowered.